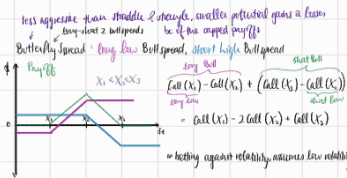


Butterfly Spread + Implied Vol

Butterfly Spread: Buy low strike bull spread, short high strike bull spread



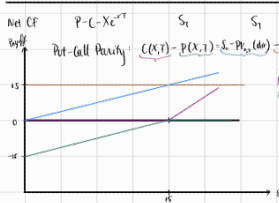
Reverse butterfly spread: bet for high volatility, but small gains if right, small losses if wrong
 → Butterfly spreads, under most assumptions, price AD converges

Dividends: price, future payoffs, risk of arbitrage opp.

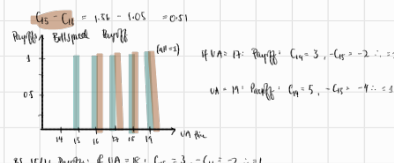
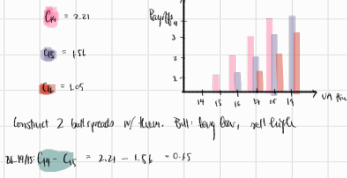
Put-Call Parity: $C(X, T) - P(X, T) = S_t - PV_{t,T}(\text{Dividends}) - e^{-rT} X$

optimal price
 risk
 dividend
 risk-free rate

Action	Today	$S_t < X$	$S_t > X$
Buy Call	-C	0	$S_t - X$
Write Put	P	$S_t - X$	0
Long PV(X) at $t=0$	$-Xe^{-rT}$	X	X



Lect ex. 3 calls for implied vol, same net 2 share. Same in currency but whole values. What do low payoffs tell you.



85.15/16 butterfly: if $VA = 15$: $C_{15} = 3, -C_{16} = -2 \therefore = 1$
 if $VA = 15$: $C_{15} = 0, -C_{16} = 0$ so both would be zero

Lect ex. Put-Call Parity

$S_t = \$110$; $C_{105} = \$17$; $X = \$105$; $B_{105} = \$5$; $r = 5\%$
 $q = 10\%$ p.a.
 $\Rightarrow 0.951818$ now

$S_t = S_0 \cdot e^{(r-q)T}$ $\hat{\delta} = 0$

$C(X, T) + X = P(X, T) + S_t e^{rT}$
 $C + Xe^{-rT} = P + S_t$
 $t=0$ $\max(S_0 e^{-rT} - X, 0) + X = \max(X - S_0 e^{-rT}, 0) + S_0 e^{rT}$
 $\max(110e^{-0.05} - 105, 0) + 105 = \max(105 - 110e^{-0.05}, 0) + 110e^{0.05}$

If $S_t > X$: $110e^{-0.05} - 105 + 105 = 0 + 110e^{0.05}$
 If $S_t < X$: $0 + 105 = 105 - 110e^{-0.05} + 110e^{0.05}$

L5: Binomial Tree

Schematic DF: $\Delta t = \frac{1}{N} \frac{Q_u}{P_u}$ Price between time for any asset

Binomial and Linear Algebra (matrix) to solve finite, but random consider: A continuum of states (functional analysis)
 Many states (discrete-time stochastic processes)
 A continuum of states (continuous-time stochastic processes)

most w/ binomial tree & then progress to each reaction concentration

Info from Prices: no arbitrage would imply: $R_t = DF_{t+1} = X$

$S_t = \frac{S_0}{2} (q S_{t+1} + (1-q) S_{t+2})$

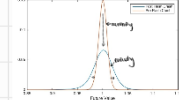
Combining 2 calls to create the butterfly spread: $85.14/15 - 85 - 15/16 = C_{15} - C_{16} = (C_{15} - C_{16})$

→ This butterfly - 14/15/16 has a payoff of 1 only when VA is equal to 15.
 → property of AD: measures → butterfly used to price AD measures

→ need to use assumptions to connect to physical paths.

→ extremely phobal context → take sophisticated look on future price movements

Risk Neutral Distribution: derived using the forward Call-Option Price Function, $C(X) = \frac{\partial^2 C(X)}{\partial X^2} = \frac{\partial^2 P(X)}{\partial X^2}$ = Risk-Neutral prob. for diffusion X

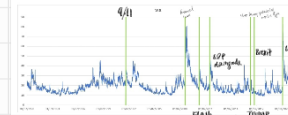


→ show people's expectations of what will happen based on what already happened

→ mean doesn't change, but the volatility increased → show people's expectations changed from new info

→ large dispersion = more volatility & uncertainty (more spread, more weight on tails)

VIX - implied volatility of S&P500 → affects expected volatility for the next month based on realized volatility (of the S&P500 index, but a proxy of market)



1) Find 2 separate

L2: Portfolio & Skewness - betting the field

2) Cash 2 carry 2 return

3) Relationship

4) CIP 2 IIP

5) Supply